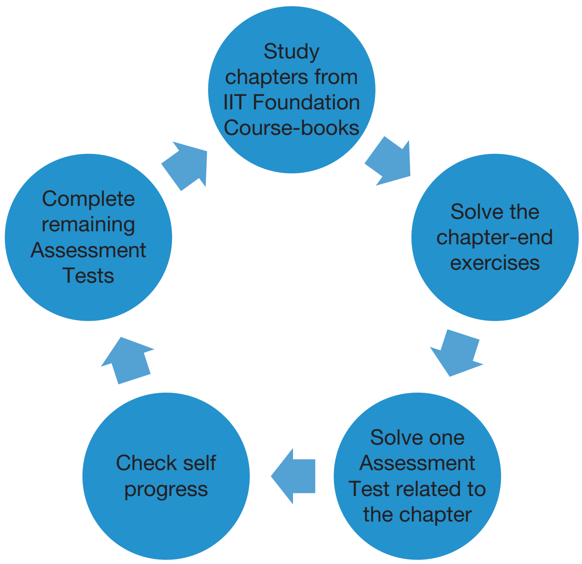
How to Use the Practice Book

Many times, students face significant challenges in answering application level questions in Physics, Chemistry and Mathematics. These Practice Books will enhance their problem-solving skill which will definitely lead to a strong subject foundation. The entire practice book series are recommended to be used alongside IIT Foundation course-books.

Students can refer the following steps while using the practice books:



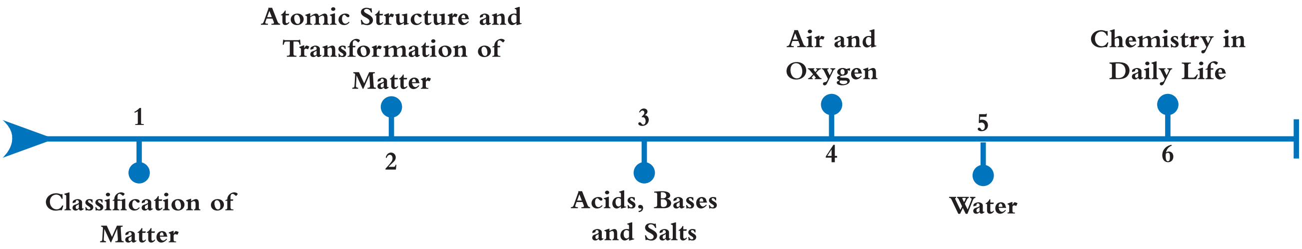
While preparing for Foundation courses, students need to learn the fundamental concepts with utmost clarity. In order to successfully complete the IIT Foundation course, one must prepare profoundly. Consistent hard work, practice and perseverance are needed throughout the year.

During any competitive examination, one must exercise clinical precision with speed since the average time available to respond to a question is hardly a minute. The aspirants should be conceptually excellent in the subject owing to the negative marking in the examination. A better practice to solve the paper would be to go for the easiest questions first and then gradually progress to the more complicated ones.

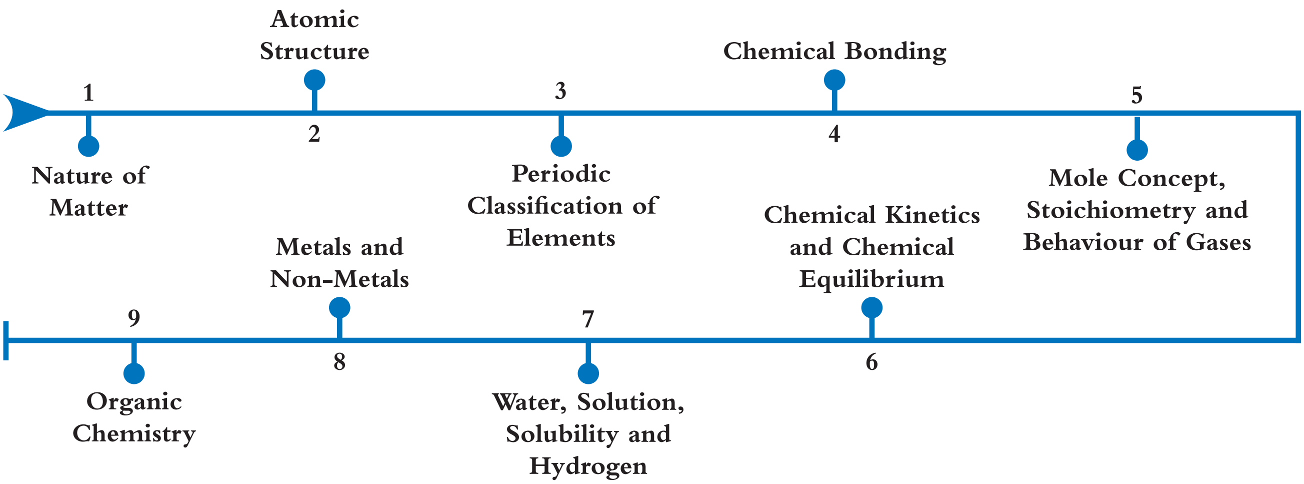
Regular practice of MCQs will assist the aspirants in preparing for the examination. In a nutshell, hard work, conceptual clarity and self-assessment are the essential ingredients to achieve success in competitive examinations. IIT Foundation course-books play an important role in understanding the concepts. Student need to read-up on all concepts/theories in a regular and systematic manner.

Course-book Chapter Flow

Class 7



Class 9



Class 8

1. Atomic Structure
2. Classification of Matter
3. Language of Chemistry and Transformation of Substances
4. Air and Oxygen
5. Water, Solution, Solubility and Hydrogen
6. Carbon and its Compounds
7. Some Important Elements and their Compounds

Class 10

1. Mole Concept, Stoichiometry and Behaviour of Gases
2. Atomic Structure
3. Periodic Table
4. Chemical Bonding
5. Acids, Bases and Salts
6. Chemical Kinetics and Equilibrium
7. Electrochemistry
8. Metallurgy
9. Industrial Chemistry
10. Organic Chemistry–I
11. Organic Chemistry–II

This page is intentionally left blank

Nature of Matter 1

Reference: *Coursebook - IIT Foundation Chemistry Class 9; Chapter - Nature of Matter; Page number - 1.1–1.17*

Assessment Test I

Time: 30 min.

Directions for questions from 1 to 15: Select the correct answer from the given options.

Space for rough work

1. In which of the following cases, phase transition will be visible?
   (a) A test tube filled with distilled water kept in a container with boiling sea water.
   (b) A test tube filled with distilled water kept in a container with boiling distilled water.
   (c) A test tube filled with sea water kept in a container with boiling distilled water.
   (d) A test tube filled with sea water kept in a container filled with the same sample of sea water which is boiling.

2. Which of the following activities does not involve the principle that evaporation causes cooling?
   (a) Cooling of water in an earthen pot.
   (b) Cooling sensation perceived while using perfume.
   (c) Fogging on spectacles after coming out from an air-conditioned car.
   (d) Usage of air cooler.

3. A metallic wire can be cut through ice even by keeping the ice block intact. Identify the term which describes this activity.
   (a) Regelation (b) Eutrophication
   (c) Osmoregulation (d) Condensation

4. Which of the following applications is based on the principle of elevation in freezing point with decrease in external pressure?
   (a) Cooking of food at high altitudes using pressure cooker.
   (b) Addition of alcohol to water used as a coolant.
   (c) Existence of ice below 0°C on mountain peaks.
   (d) Survival of aquatic animals in cold countries in winter at above 0°C.

5. Arrange the following methods of separation in the order of the following principles exploited for that purpose: Difference in boiling points; difference

Space for rough work

in solubility; selective solubility; and differential densities of solid and liquid particles.

(A) Solvent extraction (B) Centrifugation
(C) Fractional distillation (D) Fractional crystallization
(a) ACDB (b) CADB (c) CDAB (d) DCBA

6. Arrange the following in the increasing order of boiling points.
(A) Distilled water at sea level (B) Drinking water at sea level
(C) Distilled water at hill station (D) Sea water at sea level
(a) DBAC (b) CABD (c) BADC (d) CBAD

7. Which of the following is not an application of chromatography.
(a) Separation of components of blood.
(b) Identification of impurity in a mixture.
(c) Identification of food adulteration.
(d) Doping tests conducted in sports events.

8. **Assertion (A):** Ethyl alcohol and water can be separated by separating funnel.
Reason (R): Ethyl alcohol and water differ with respect to their boiling points.
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true, but R is not the correct explanation of A.
(c) A is true and R is false.
(d) A is false and R is true.

9. Identify the odd one among the following.
(a) Iodine + sulphur (b) Ammonium chloride + sand
(c) Sulphur + sand (d) Naphthalene + $BaSO_4$

10. Two liquids 'X' and 'Y' can be separated by a separating funnel. Y and water can also be separated by the same method. Which of the following conclusions can be inferred on the basis of the above information?
(i) X and Y are miscible liquids.
(ii) A mixture of X and water can be separated by fractional distillation.
(iii) X and Y are immiscible liquids.
(iv) A mixture of Y and water can also be separated by fractional distillation.
(a) (i) and (ii) (b) (iii) and (iv)
(c) (ii) and (iii) (d) (ii), (iii) and (iv)

Space for rough work

11. Which of the following statements is false regarding fractional crystallization?

(a) A hot saturated solution of mixture is taken.

(b) Crystals of the less soluble component get separated first.

(c) The entire amount of component can be separated by cooling up to 0°C.

(d) A more soluble component is left behind in the solution.

12. Match the statements of Column A with those of Column B.

Column A	Column B
(i) Fractional evaporation	(A) H_2 and NH_3
(ii) Dissolution in KOH	(B) H_2 and Ar
(iii) Preferential liquefaction	(C) NH_3 and O_2
(iv) Diffusion	(D) N_2 and O_2
(v) Dissolution in water	(E) SO_2 and N_2

(a) (i) → (B); (ii) → (E); (iii) → (A); (iv) → (D); (v) → (C)

(b) (i) → (B); (ii) → (E); (iii) → (D); (iv) → (C); (v) → (A)

(c) (i) → (D); (ii) → (E); (iii) → (A); (iv) → (B); (v) → (C)

(d) (i) → (D); (ii) → (A); (iii) → (B); (iv) → (E); (v) → (C)

13. Which of the following mixtures can be separated by fractional crystallization?

(a) $P + NaNO_3$ (b) $NaNO_3 + NaCl$

(c) $KNO_3 + C$ (d) $NaCl + S$

14. **Assertion (A):** A mixture of SO_2 and water can be separated by heating.

Reason (R): SO_2 is highly soluble in water.

(a) Both A and R are true and R is the correct explanation of A.

(b) Both A and R are true, but R is not the correct explanation of A.

(c) A is true and R is false.

(d) A is false and R is true.

15. Which of the following methods of separation is used in the refining of petroleum?

(a) Fractional evaporation (b) Fractional distillation

(c) Fractional crystallization (d) Solvent extraction

Assessment Test II

Time: 30 min.

Space for rough work

Directions for questions from 1 to 15: Select the correct answer from the given options.

1. In which of the following cases, is phase transition not at all possible?
   (i) Test tube with sea water in ice.
   (ii) Test tube with sea water in freezing mixture.
   (iii) Test tube with distilled water in ice.
   (iv) Test tube with distilled water in freezing mixture.
   (a) (i) and (iii) (b) (ii) and (iv)
   (c) (ii) and (iii) (d) (i) and (iv)

2. Identify the true statements regarding evaporation from the following:
   (i) Rate of evaporation increases with humidity.
   (ii) Evaporation is a surface phenomenon.
   (iii) Evaporation results in decrease in the temperature of the surroundings.
   (iv) Rate of evaporation is independent of external pressure.
   (a) (i) and (iii) (b) (ii) and (iv)
   (c) (ii) and (iii) (d) (i) and (iv)

3. Identify the odd one among the following with respect to the principle involved.
   (a) Skating on ice.
   (b) Formation of water droplets on the surface of a tumbler filled with ice.
   (c) Fogging of exhaled air in winter morning.
   (d) Formation of dew.

4. **Assertion (A):** Sea water freezes below 0°C.
   Reason (R): Freezing point increases due to the addition of impurities.
   (a) Both A and R are true and R is the correct explanation of A.
   (b) Both A and R are true, but R is not the correct explanation of A.
   (c) A is true and R is false.
   (d) A is false and R is true.

5. Arrange the following mixtures in the order of separation methods given below:
   Gravity separation, centrifugation, fractional distillation, distillation, separating funnel
   (A) KNO_3 + water (B) Petrol + alcohol
   (C) Water + alcohol (D) Sand + finely powdered chalk
   (E) Blood

Space for rough work

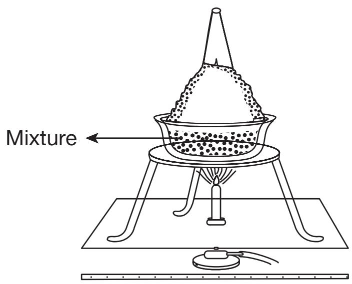
(a) DECAB (b) DCAEB
(c) CADEB (d) CEABD

6. Arrange the following in the decreasing order of cooking time.
(A) Cooking in a vessel covered with a lid at a hill station.
(B) Cooking in pressure cooker at sea level.
(C) Cooking in open vessel at hill station.
(D) Cooking in pressure cooker at hill station.
(a) ADCB (b) CDBA
(c) DCAB (d) CADB

7. Identify the term which is not related to the method of separation of leaf pigments.
(a) Distillate (b) Stationary phase
(c) Mobile phase (d) Adsorption

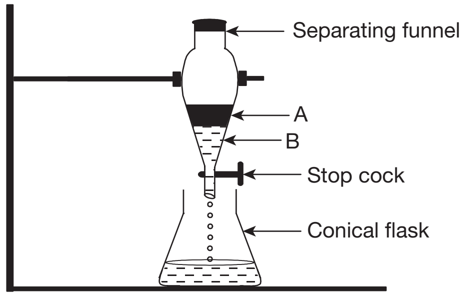
8. A mixture of two liquids 'X' and 'Y' is taken. The boiling points of X and Y are 80°C and 110°C, respectively. Which of the following conclusions would be the correct experimental result?
(a) X is the distillate and is collected in the receiver flask.
(b) Y is the distillate and is collected in the receiver flask.
(c) X is the distillate and is left in the distillation flask.
(d) Y is the distillate and is left in the distillation flask.

9. Which of the following mixtures can be separated by the method represented by the following figure?



(a) Sulphur and sand
(b) Potassium nitrate and potassium chloride
(c) Iodine and ammonium chloride
(d) Ammonium chloride and sand

10. The following figure represents a separation method. By observing the figure, identify the true statements among the following.



(A) A and B are immiscible liquids.
(B) A and B form a homogeneous mixture.
(C) Liquid A is heavier than liquid B.
(D) Liquid B is denser than liquid A.

(a) (A) and (B) (b) (C) and (D)
(c) (A) and (D) (d) (B) and (D)

11. When a hot, saturated solution of two substances 'X' and 'Y' at 80°C is cooled slowly, X first crystallizes and settles down at 30°C, whereas Y remains in the solution. Identify the true statement which complies with the above experimental result.

(a) X is more soluble than Y at 80°C.
(b) X is more soluble than Y at 30°C.
(c) Solubility of Y at 30°C is more than that of X.
(d) Solubility of Y is more than that of X from 80°C to 30°C.

12. Match the statements of Column A with those of Column B.

Column A	Column B
(i) Preferential liquefaction	(A) Difference in molecular masses
(ii) Fractional evaporation	(B) Preferential solubility of a gas in a solvent
(iii) Diffusion	(C) Difference in critical temperatures
(iv) Dissolution in a solvent	(D) Difference in boiling points

(a) (i) → (C), (ii) → (D); (iii) → (A); (iv) → (B)
(b) (i) → (C); (ii) → (D); (iii) → (B); (iv) → (C)
(c) (i) → (C); (ii) → (A); (iii) → (D); (iv) → (B)
(d) (i) → (B); (ii) → (D); (iii) → (A); (iv) → (C)